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ORIGINAL RESEARCH

A Survey on Knowledge, Attitude, and Practice of Virtual Schooling in Dental Undergraduate and Postgraduate Education

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Abstract :

The purpose of this study was to assess the knowledge, attitude, and practice (KAP) of Indian undergraduate and postgraduate dental students towards virtual learning and to better understand the implementation of this learning strategy. A complement to provide tools to further your dental training. A closed-ended, multiple-choice, cross-sectional online questionnaire was distributed nationwide among her dental and postgraduate students across India. Questions focused on virtual platform selection, learning methods, class duration, mental health, advantages and disadvantages, and approaches to virtual education as a complementary tool in the future. The survey was distributed on various social platforms in the form of web-based links and quick response. Descriptive statistics were performed to compile the data. A total of 2,130 students (1,318 undergraduates and 812 graduate students) participated in the study's survey found that 81.1% of undergraduate students and 92.5% of graduate students were satisfied with her virtual learning. The most commonly used platforms for online education were Zoom and Google Meet. The most common form of virtual learning was webinars conducted by expert faculty. Easy and convenient access and time flexibility are the most common advantages, while internet connectivity problems and negative psychological effects are some common disadvantages. The concept of blended learning was most preferred by students. Undergraduate and postgraduate students showed good knowledge and a positive attitude to virtual schooling and preferred a hybrid model of instruction.

Keywords : Virtual learning, Dental education, Medical education

Introduction :

The pandemic caused by coronavirus disease 2019 (COVID-19) has forced lockdowns everywhere and the medical community has urged everyone to practice social distancing to prevent the spread of the disease.

This caused riots in many sectors of society and affected the education sector. Changes in training methods were needed to provide uninterrupted training and support the academic development of dental students. In some areas of dentistry, traditional face-to-face learning is being replaced by virtual learning methods.

The rapid development of computer technology and the growth of the Internet infrastructure over the years have led to the rapid development of online education. Expanded access to computers, faster network connections, and technology have led to increased adoption in many fields, and dental education is no exception. Social media platforms that provide educational tools through smart devices in the form of apps such as Facebook, YouTube, Twitter, Zoom, Instagram, Skype, etc., are popular among younger generations to collaborate, connect and communicate at a global level [1].

This early adoption of his by the younger generation paved the way for the field of learning through virtual education. Virtual Learning mainly consists of webinars, demonstration videos, case studies, and his discussions on social media platforms such as Zoom, Google Meet, Google Classroom, and Skype. A previous literature study. The academic anatomy community suggested that there may be value in integrating social media into blended learning approaches in anatomy programs. This is because the majority of learners preferred electronic (e) learning over traditional classroom learning due to accessibility, time constraints, ease of understanding, and more visual representation [2].

This learning method has also been adopted by medical and dental education students to keep up with the ever-increasing number of new and updated topics. Although there were many advantages, disadvantages included difficulties such as access to technology for all students and universities, prolonged attention spans in front of computer screens, student easily distracted, and inappropriate interpersonal communication. There was also, impacts on peers, teachers, and mental health. Virtual learning alone may not be sufficient to provide comprehensive dental education. However, integrating virtual learning with traditional classroom learning can improve student learning [3].

Understanding the advantages, disadvantages, obstacles, and challenges of virtual education compared to traditional learning through the perceptions of undergraduate and graduate students will help create a structured protocol for future her dental education helpful. Therefore, the purpose of this study was to evaluate dental students' and graduate students' knowledge, attitudes, and perceptions (KAPs) and approaches to dental virtual to assess dental virtual as a complementary tool in dental education [4]. The idea was to conduct a nationwide survey, education system.

Materials and methods:

Participant selection and description:

A national cross-sectional online survey was conducted among dental students across India. All dental students, interns, and doctoral students participated in the study.

Survey:

The survey consists of closed-ended, multiple-choice questions created using the Smart Survey online application and can be found in the Appendix section. All data is securely stored on the sub processor.

The questionnaire used in this study was developed and validated based on a pilot study. A preliminary pilot study was conducted with 20 of her undergraduate students and 20 of her graduate students who underwent virtual learning [5]. An email containing a link and QR code (Quick Response) via WhatsApp was used to distribute the pilot survey. The questionnaire was validated and feedback results were taken into account when creating the final questionnaire. Survey consists of 22 closed-ended multiple-choice questions for undergraduate students and 19 for graduate students. Basic demographic data were collected for students and their training year. The questions formulated focused on the choice of social platforms, trends towards learning methods, advantages and disadvantages of virtual learning, duration of online courses, mental health of students, and perception of virtual education as an auxiliary tool of society. I guessed it, Future dental education [2,4].

The survey was worded so that students new to virtual learning were not forced to answer all questions, and were ultimately forwarded to record their feelings about virtual education. Prior consent was obtained from all participants, and those who did not wish to consent were invited to exit the study. Participants were not asked personal questions, and her email containing a link explaining the intent of the study was distributed to students. Validated questionnaire was used for the final questionnaire [6]. The final validated questionnaire, created using hypertext markup language (HTML), was provided to potential respondents via a web server. This link was shared to all undergraduate and postgraduate students across the country via email, WhatsApp, Instagram, Facebook and other platforms. It was distributed to the expert group as a personal email and messages. Exclusion criteria consisted of students who participated in the pilot study and members of the survey design team. Three reminder messages were sent within a week.

Survey links are now disabled after the last reminder his message to avoid duplicate answers. The anonymity of the study was ensured because Internet Protocol (IP) addresses were collected and the researchers had no way to identify the participants.

Statistical analysis:

All collected data were analyzed using IBM SPSS Statistics for Windows version 23.0 (2015 release, IBM Corp., Armonk, NY, USA). Descriptive statistics were used to present the results of the study.

Results:

A total of 2,130 students (1,318 undergraduates and 812 graduate students) participated in the survey, of which 16 did not complete the survey. The response rate for the survey was an overwhelming 99%. A total of 1,302 participants (74.5% male, 25.5% male) were undergraduate students, and 812 (73.2% male, 26.8% male) were graduate students. In this study, 92.5% of graduate students and 81.1% of undergraduate students were familiar with virtual learning methods. Students who denied receiving any form of online learning were asked more general questions and did not face questions about virtual learning [7].

Undergraduates:

Second-year undergraduates made up the majority of the sample (29.1%). Other students in her grade were represented as well. The majority of undergraduate students (64.1%) used their mobile phones for their web-based learning through various platforms. A significant proportion of students used the Zoom platform (78.22%) for their undergraduate dental education, followed by Google Meet (30.21%) and WhatsApp video platform (16.57%). Publicly accessible video/audio communication platforms were used by (29.9%) students, while students enrolled in educational institutions who expressed security and privacy concerns (24.3%) I was using an educational institution's platform. 45.74% of the students attended classes 6 days a week, days, while some students attended classes 3 to 5 days a week [8]. Faculty-led online seminars/lectures were the most common learning format, with the most preferred duration ranging from 45 to 60 minutes. Virtual education as an instructional method used in educational institutions includes webinars or lectures (72%), homework (54.5%), and online tests/exams (43.4%) [9].

The learning methods most frequently selected by students were webinars (60%), clinical demonstrations (55.8%), case scenarios (44.8%), activity learning (44.3%), and group discussions (43.5%). Activity-based learning methods such as flowcharts (69.5%), mind maps (50.7%), and quizzes (49%) were preferred by many of her students. The majority of students rated the interaction between students and instructors as equal (34.5%) or good (33.7%), while only 21.9% of students perceived the interaction as poor [10]. Traditional classroom learning ideas selected. 48% of respondents supported virtual learning only, only 6.54% supported virtual learning, but 45.5% of students said it would be more beneficial to combine the two learning methods Key aspects of virtual education The advantages and disadvantages are shown in (table 1,2).

S. no.	Advantages	Response
1	Flexibility of time	60.45%
2	Ease of access (using computer or smartphone)	55.15%
3	Access from any location	65.05%
4	Affordable	42.70%
5	Access to faculty without any limitation	30.65%
6	Multidimensional possibility to enhance knowledge through online sources	29.03%
7	Ease of assessment	29.26%
8	Environment-friendly	30.80%
9	Faster access to and sharing of educational resources	44.24%
10	Others (please specify)	1.54%

Table 1: Survey questions and responses pertaining to the advantages of virtual schooling.

S. no.	Disadvantages	Response
1	Internet connectivity issues	88.16%
2	Cost of internet connection	31.07%
3	Cost of devices	21.58%
4	Internet security issues	52.03%
5	Lack of face-to-face interaction with faculty	53.76%
6	Lack of interaction with patients	65.23%
7	Lack of interaction with classmates	65.23%
8	Laziness and procrastination	44.88%
9	Need for self-discipline	29.22%
10	Poor pre-clinical and practical orientation	45.99%
11	Feeling of isolation, lack of social contact	17.02%
12	Impact on physical and mental health	45.50%
13	Others (please specify)	0.74%

Table 2: Survey questions and responses pertaining to the disadvantages of virtual schooling.

The biggest benefits of virtual learning were convenience of access from locations (65.05%) and schedule flexibility (60.1%), followed by ease of access (55.2%). The most common barrier to virtual education was internet connectivity issues (85.7%), followed by lack of patient interaction (56.91%). Other common issues faced by students were internet security issues (41.6%) and negative health effects (49.6%). His 50.4% of survey respondents believe that virtual learning impairs clinical understanding. The idea of introducing it in future dental education was not supported by 50.1% of students. While a significant number of students (43.6%) believed it had a negative impact on their overall health, 72.9% agreed about its usefulness during the pandemic [11].

PhD Students:

The survey included first, second, and third year doctoral students from all areas of dentistry, including orthodontists. More graduate students used laptops (65.5%), cell phones (26.3%), tablets (5.6%), and desktop computers (2.7%) to take online courses.

The teaching mode provided by web-based applications for learning is Zoom application (85.7%), followed by Google Meet (29.2%), Cisco Webex (26.4%), GoTo Webinar (22%), and Facebook Live. (17.5%). Virtual learning through faculty-provided webinars took 60 minutes, and magazine viewing took 45 minutes.

Various other types of virtual learning are listed in Table 3. The school days were held six days a week for 33.3% of students, and the remaining students had classes three to five days a week. 46.3% of students reported that their interactions with instructors in virtual sessions were the same as in-person sessions, and 33.2% of students felt that interactions were better; A minority (20.53%) said they felt this way [12]. The interaction was much worse. A majority of 58.8% of students felt that both virtual learning and in-person instruction were beneficial to them, but 35.6% of graduate students still supported the idea of in-person instruction.

Convenient access from anywhere was the biggest advantage of this educational model, according to (72.9%) of graduate students. Other benefits include scheduling flexibility (62.8%), ease of access (58.8%), faster resource sharing (56.5%), and affordability (45.7%). In this survey, 88.16% of students, or students, reported that internet connectivity issues pose challenges to continuing their online education. Other disadvantages targeted by many students were lack of interaction with patients (65.23%) and teachers (53.76%).

The idea of holding conferences on a virtual platform was supported by 47.5% of PhD students. A large number of paper presentations (58%) and poster presentations on virtual platforms (62%) were also selected. Over 89.5% of respondents felt that the virtualization of education was very helpful during the lockdown period. Approximately 47.5% of students agreed that using virtual platforms as a complement to traditional methods rather than as the primary form of education is a good strategy [4].

Discussion:

The purpose of this study was to assess the perceptions and knowledge of undergraduate and graduate students about virtual learning as a complement to traditional methods in dental education found that 95.7% of the students found e-learning attractive, similar to the students in this study. The designed survey highlighted various aspects of virtual learning from the students' perspective, and the majority of participants were female (74.5% undergraduates, postgraduates) as many women are undergoing dental training. (73.2%).

The majority of student's report using Zoom, WhatsApp video calls, Google Meet, and YouTube Live, with an astonishing 81.1% of undergraduates and 92.5% of graduate students using these platforms for educational purposes. This is similar to previous research [3]. In this study, the majority of the students chose the Zoom option, followed by Google Meet and WhatsApp video calls for online classes, with a minority using other platforms.

Although many students used mobile phones for virtual courses, less than one-third of undergraduates reported using a tablet or laptop and used a desktop computer for learning. The least number of students reported this at which is consistent with the following results. Other studies. The attitudes of the students towards the use of mobile phones as a medium for dental education were very positive due to the availability and accessibility of inexpensive phones and mobile data packages [9]. Approximately 24.5 of the D44 students preferred secure video conferencing platforms as a form of dental education. Similar, there is also a study by Lin et al. found high usage and preference for smart phones among students, which is similar to the present study. This includes designing content for future dental educators to be mobile-friendly as well. For activity-based learning, Mind Mapping, Flowcharts, and Quizzes were at the top of the list. Similar to the present study, student prioritized quizzes and homework.

Webinars conducted by faculty for students ranged in duration from 45 to 60 minutes, which was also the preferred option by students. The number and duration of online courses and the number of enrolled students have increased significantly. Interaction between students and teachers was very low. 35.5% of students and 33.2% of graduate students were convinced, and the rest were of the opinion that classroom learning improves the interaction between other students and teachers. Blended learning was supported by 45.55% of undergraduate students and 58.80% of graduate students [2]. These results were consistent with those of the study. E-learning is considered by the medical community to be less effective than face-to-face education in terms of improving clinical skills mentioned, which was similar to his current study?

The basic challenge faced by students was internet connectivity issues (85.79%). Unreliable internet connection is one of the biggest problems. Higher bandwidth connections are not available or affordable to all students [8]. This was one of the most common difficulties with virtual school. In addition to connectivity issues, internet security issues were also a concern for 48.54% of his respondents, which is on par with other surveys.

Lack of interaction with teachers, patients and classmates is one of the main problems, students said. The lack of direct clinical evaluation for patients may bias and compromise clinical understanding of the subjects (50.4%). Another group of negative aspects of virtual learning that students agreed were negative attitudes, procrastination (44.9%) affecting self-control (29.2%), and the impact on overall physical and mental health. There were negative effects (45.5 people). %), leading to greater efficiency. In an attempt to assess the relationship between mental health and virtual learning While 50% of students experienced no negative mental health effects of students reported mental health issues. Students reported experiencing physical and mental disorientation, which negatively impacted their overall health and education [7]. Studies have reached similar conclusions, stating that online learning has a negative impact on mental health.

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Conflict of interest :

Author declares there is no conflict of interest.

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